BAO CAO THUC NGHIEM LAB01

Run ID: 20261105_013738_9a4527
Trang thai: SUCCESS
Thoi gian bat dau: 2026-11-05T01:37:38+07:00
Thoi gian ket thuc: 2026-11-05T01:38:40+07:00

1. TIEN TRINH THUC HIEN

STT	Buoc thuc hien	Trang thai
1	EDA	SUCCESS
2	PREPROCESSING	SUCCESS
3	TRAINING - Logistic Regression	SUCCESS
4	TRAINING - Naive Bayes	SUCCESS
5	TRAINING - SVM	SUCCESS
6	TRAINING	SUCCESS
7	EVALUATION - Logistic Regression - Train	SUCCESS
8	EVALUATION - Logistic Regression - Test	SUCCESS
9	EVALUATION - Naive Bayes - Train	SUCCESS
10	EVALUATION - Naive Bayes - Test	SUCCESS
11	EVALUATION - SVM - Train	SUCCESS
12	EVALUATION - SVM - Test	SUCCESS
13	EVALUATION	SUCCESS
14	PIPELINE	SUCCESS

2. KET QUA DANH GIA MO HINH

Run_ID	Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
20261105_013738_9a4527	Logistic Regression	Train	0.95354729729729973	0.90078328981723323	0.95303867403314912	0.92617449664429953	0.95303867403314912	0.04622871046228710 4613	0.97163614382116 13	3136	152	68	1380
20261105_013738_9a4527	Logistic Regression	Test	0.93158783783783 78	0.84691358024691 35	0.94751381215469 61	0.89439374185136 89	0.94751381215469 61	0.07542579075425 791	0.95967926227635 06	760	62	19	343
20261105_013738_9a4527	Naive Bayes	Train	0.99619932432432 43	0.98972602739726 02	0.99792817679558 01	0.99381017881705 64	0.99792817679558 01	0.00456204379562 0438	0.99995904242448 69	3273	15	3	1445
20261105_013738_9a4527	Naive Bayes	Test	0.97297297297297 3	0.96348314606741 57	0.94751381215469 61	0.95543175487465 17	0.94751381215469 61	0.01581508515815 0853	0.99354760656531 03	809	13	19	343

Run_ID	Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
20261105_013738_9a4527	SVM	Train	0.71938344594594 59	0.88888888888888 888	0.09392265193370165	0.16989381636477202	0.09392265193370165	0.00517031630170316 3	0.97475858134720 59	3271	17	1312	136
20261105_013738_9a4527	SVM	Test	0.71875	0.89189189189189 19	0.09116022099447514	0.16541353383458646	0.09116022099447514	0.00486618004866 1801	0.96472691589036 31	818	4	329	33

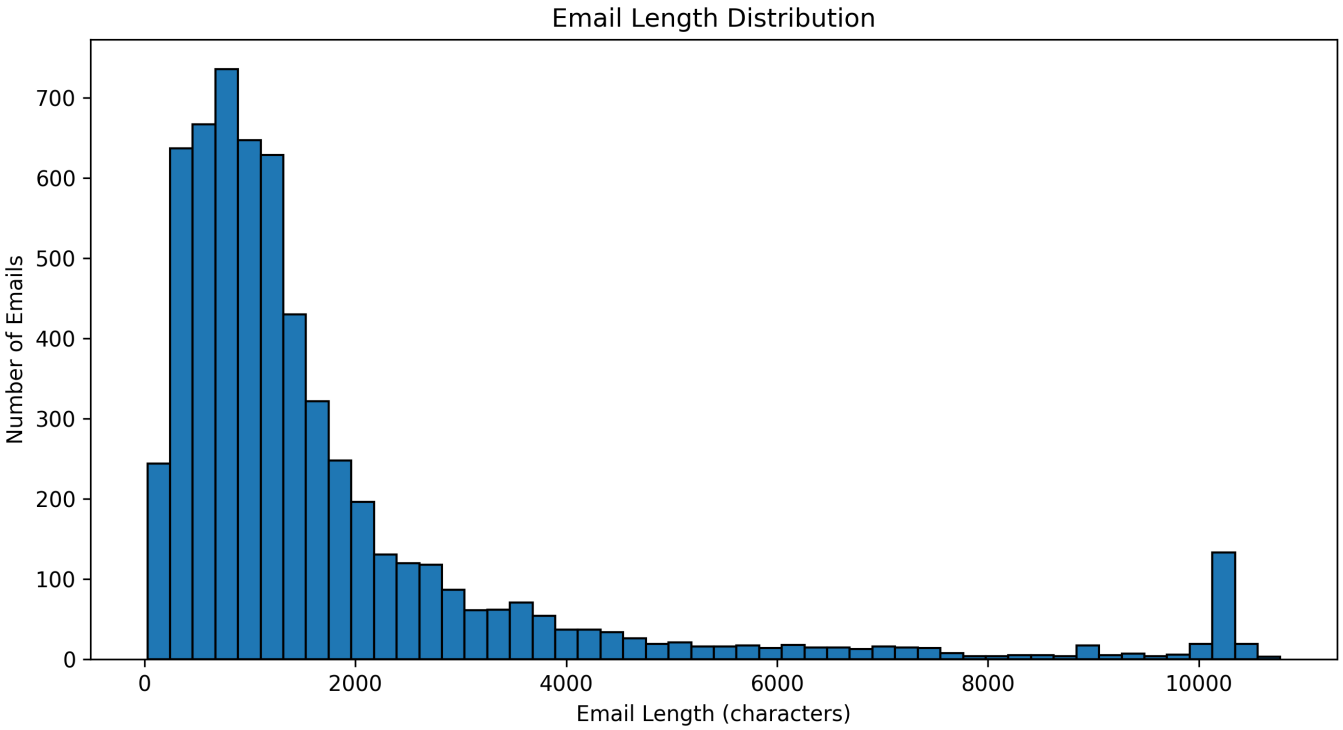
3. MO HINH TOT NHAT

Mo hinh: Naive Bayes
F1-score: 0.9554317548746517

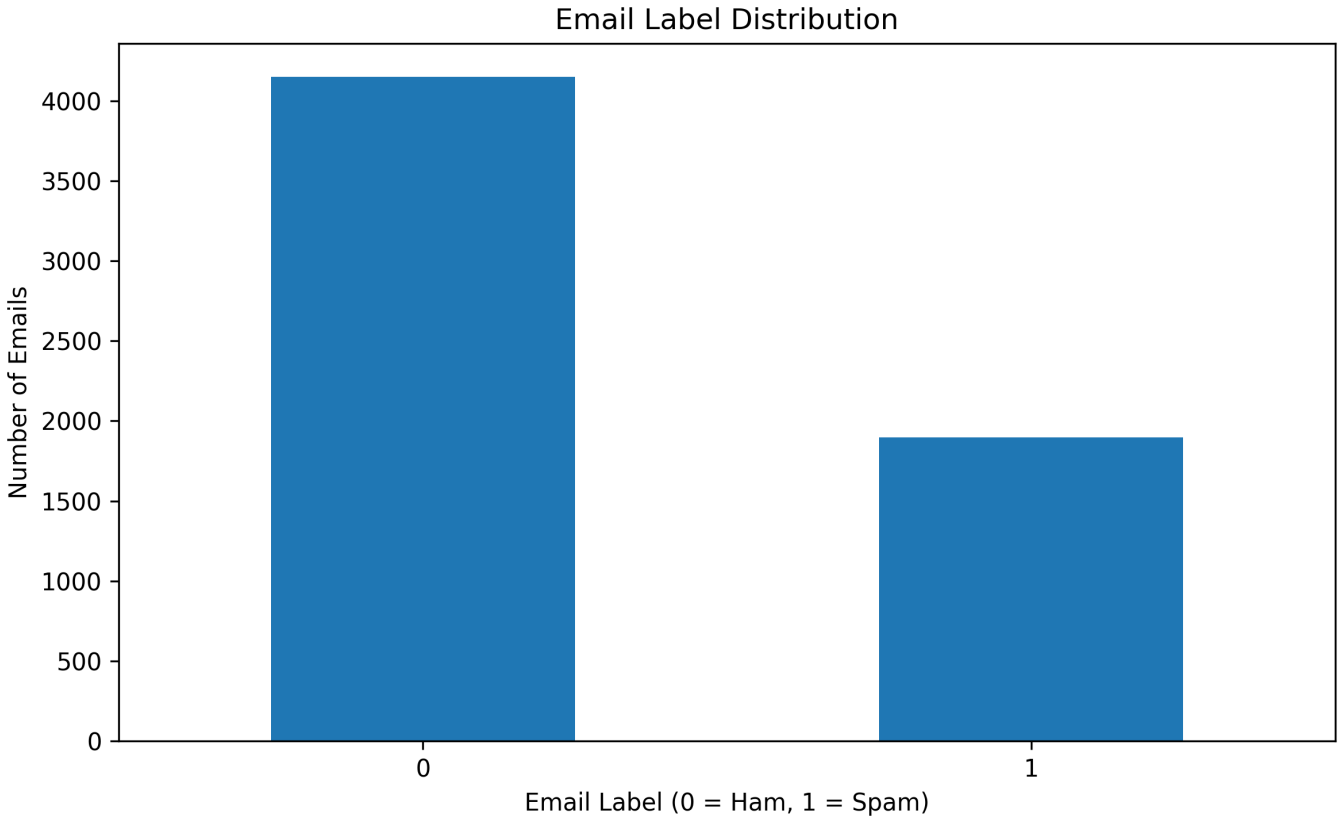
4. BIEU DO KET QUA

EDA

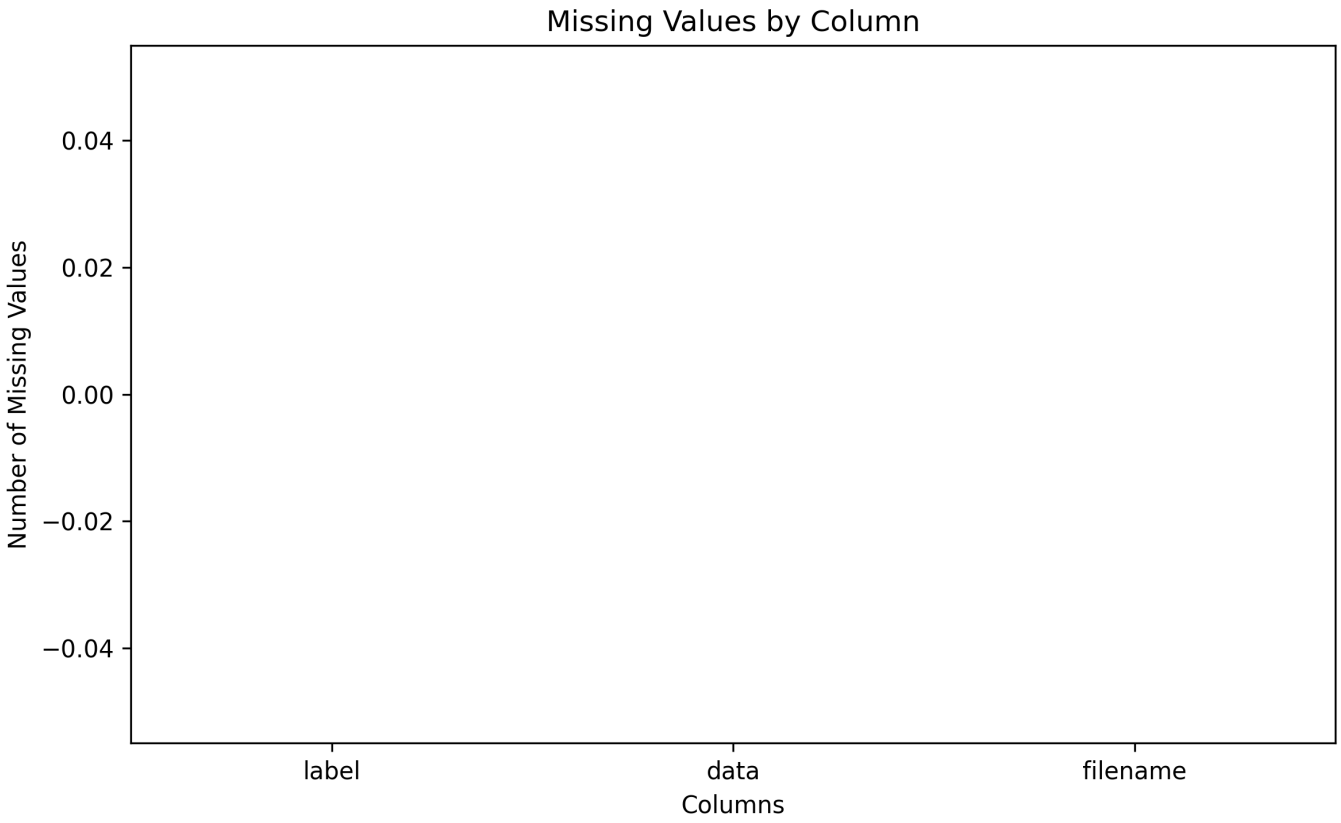
email_length_distribution



label_distribution

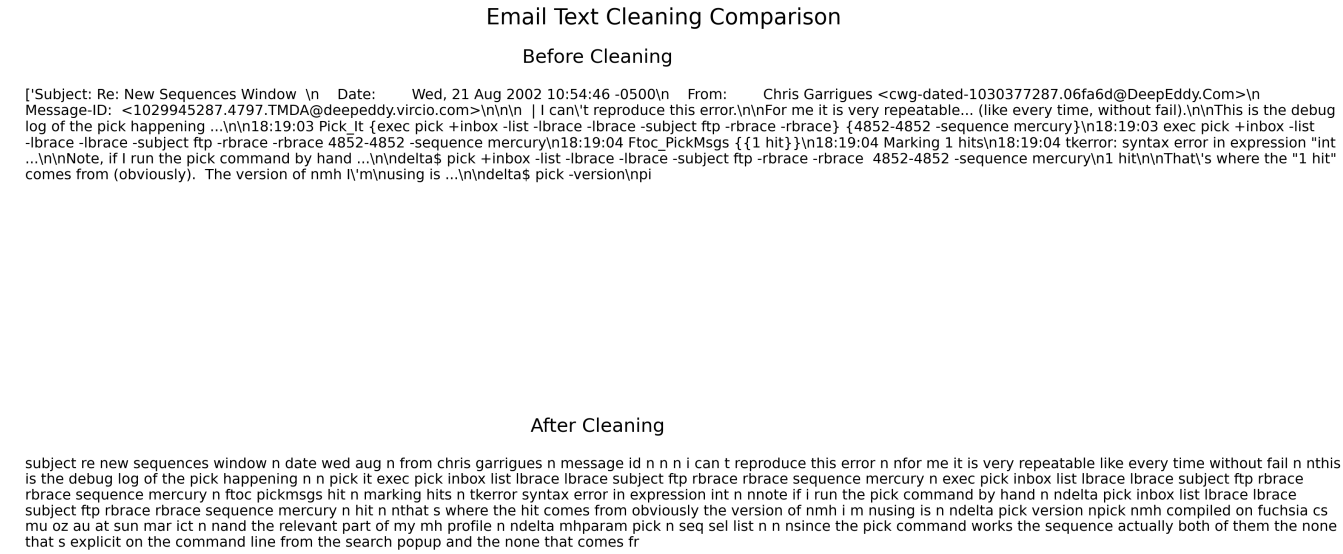


missing_values

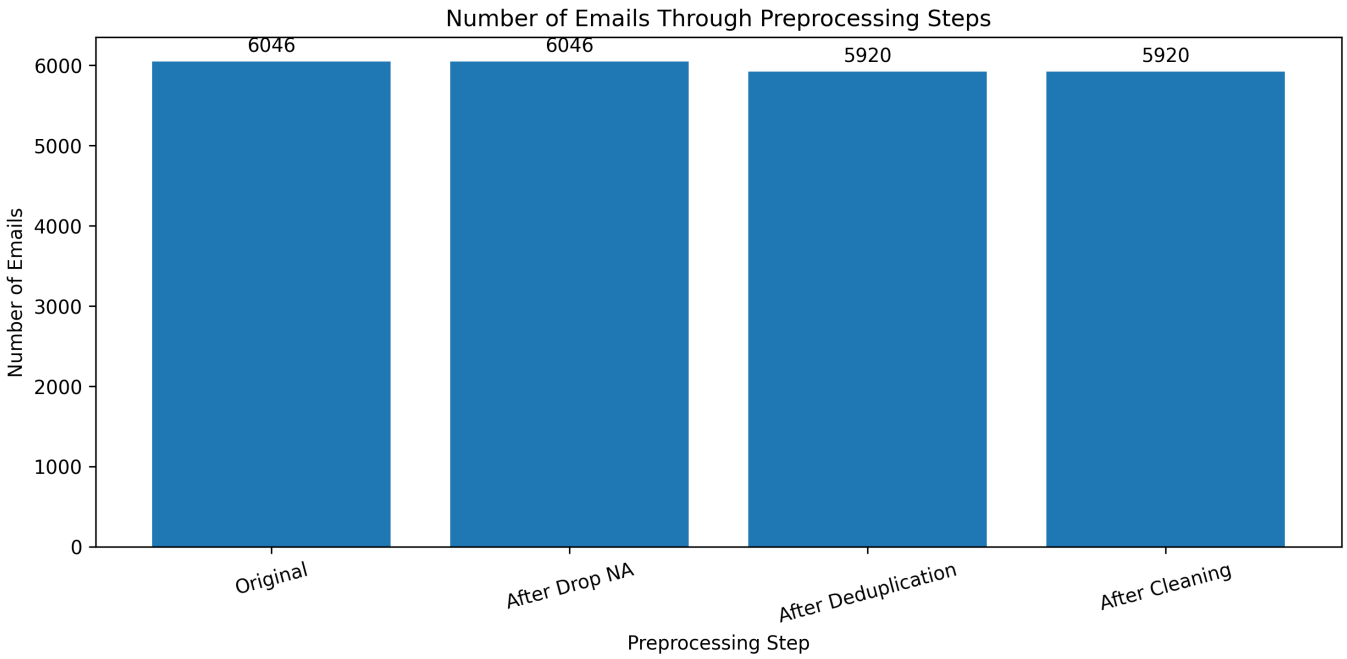


PREPROCESSING

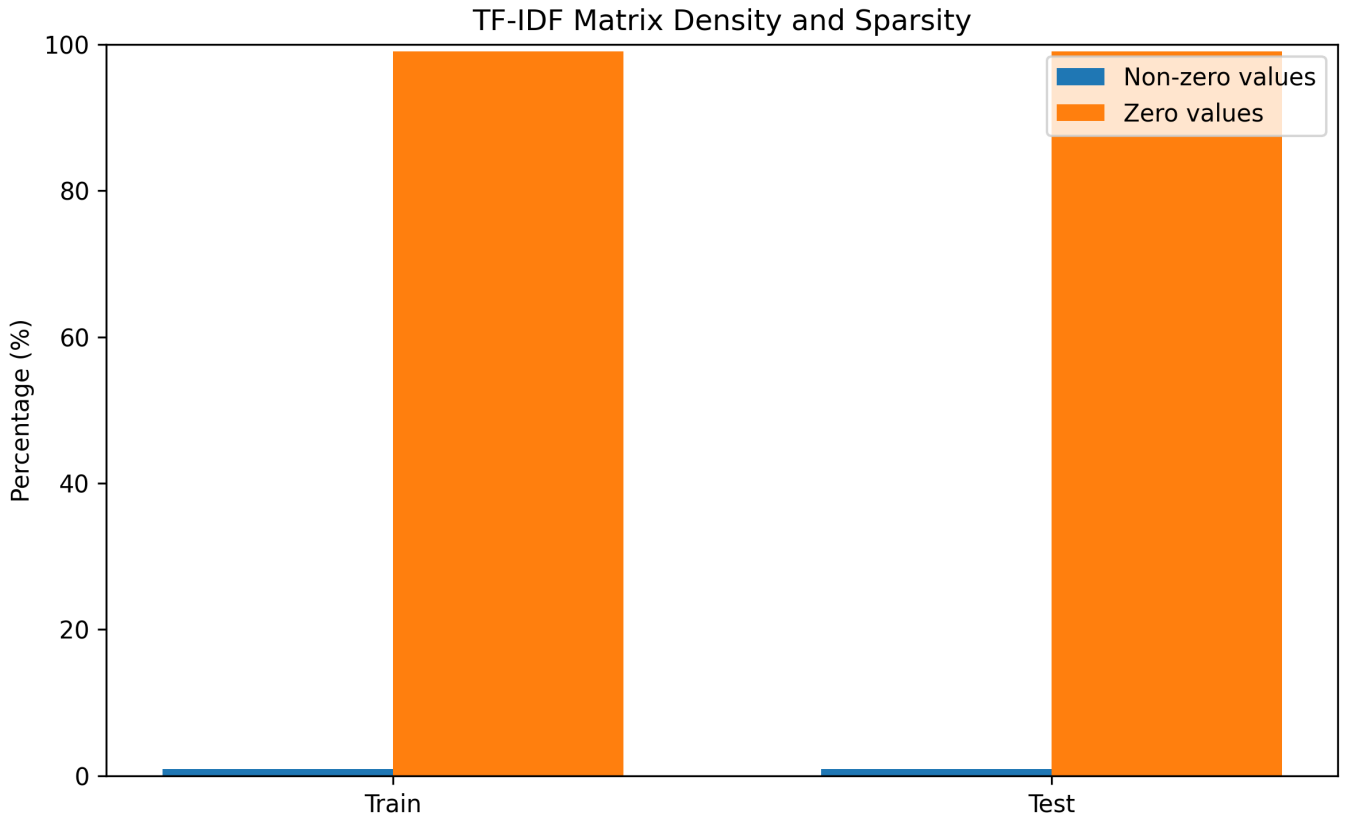
cleaning_comparison



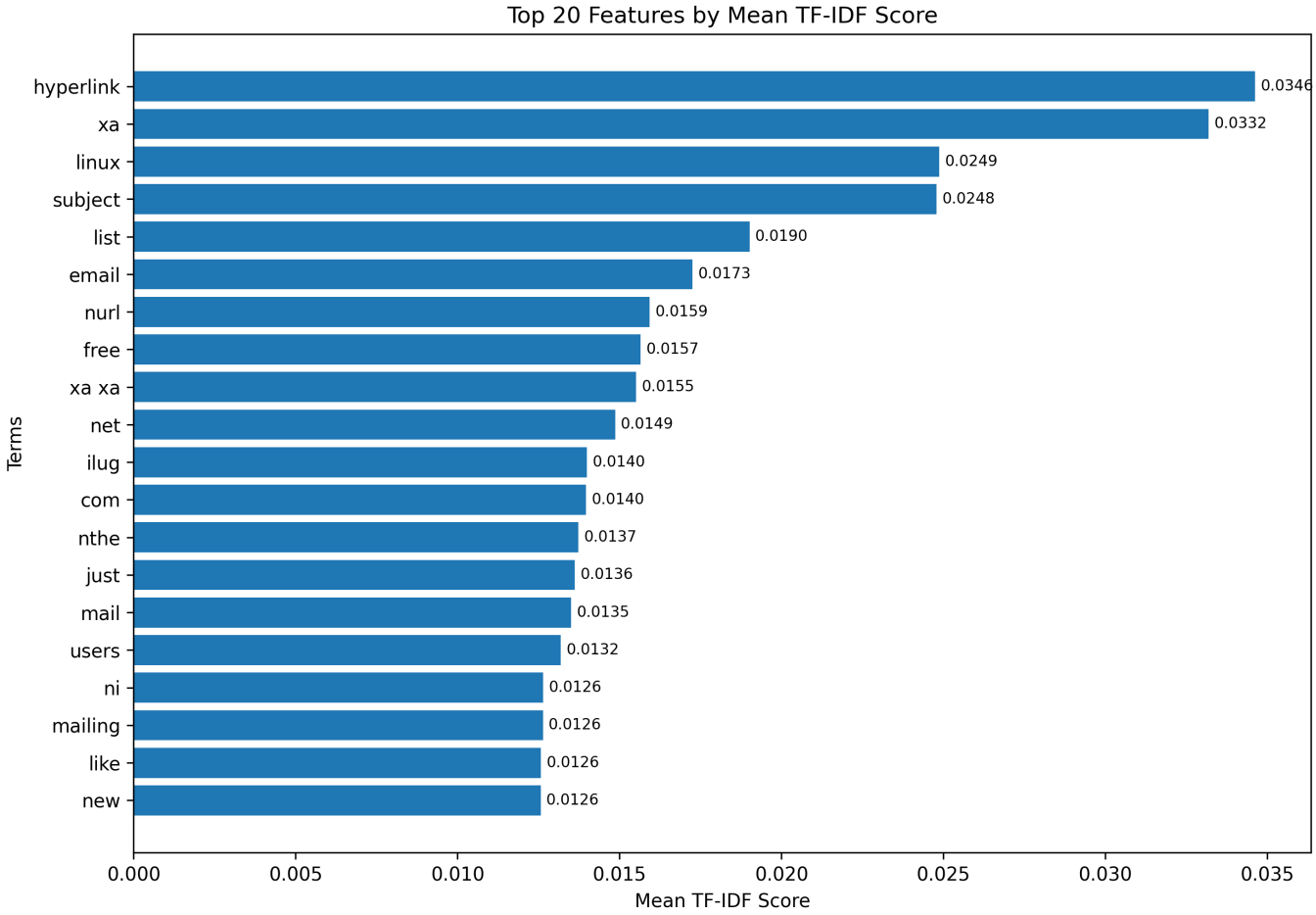
preprocessing_steps



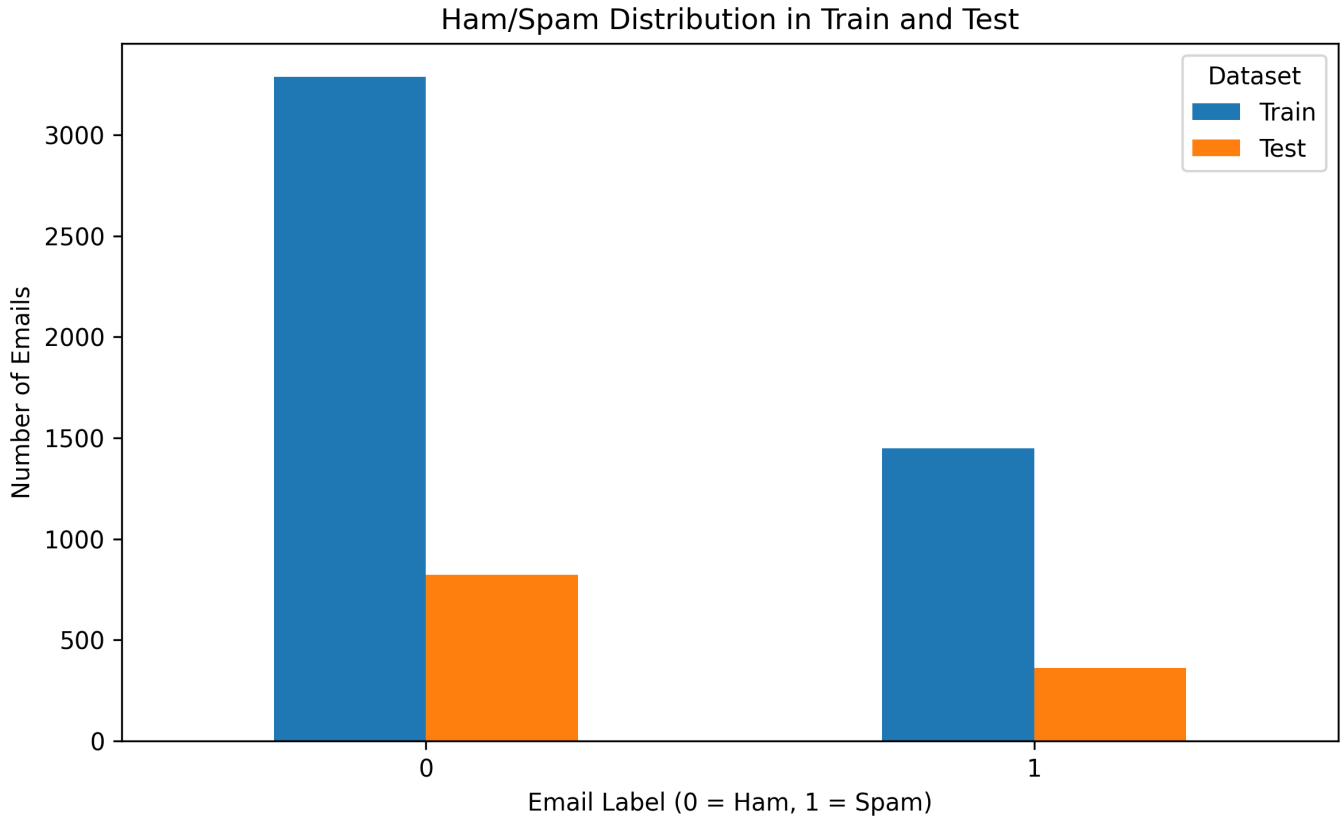
tfidf_sparsity



tfidf_top_features



train_test_distribution



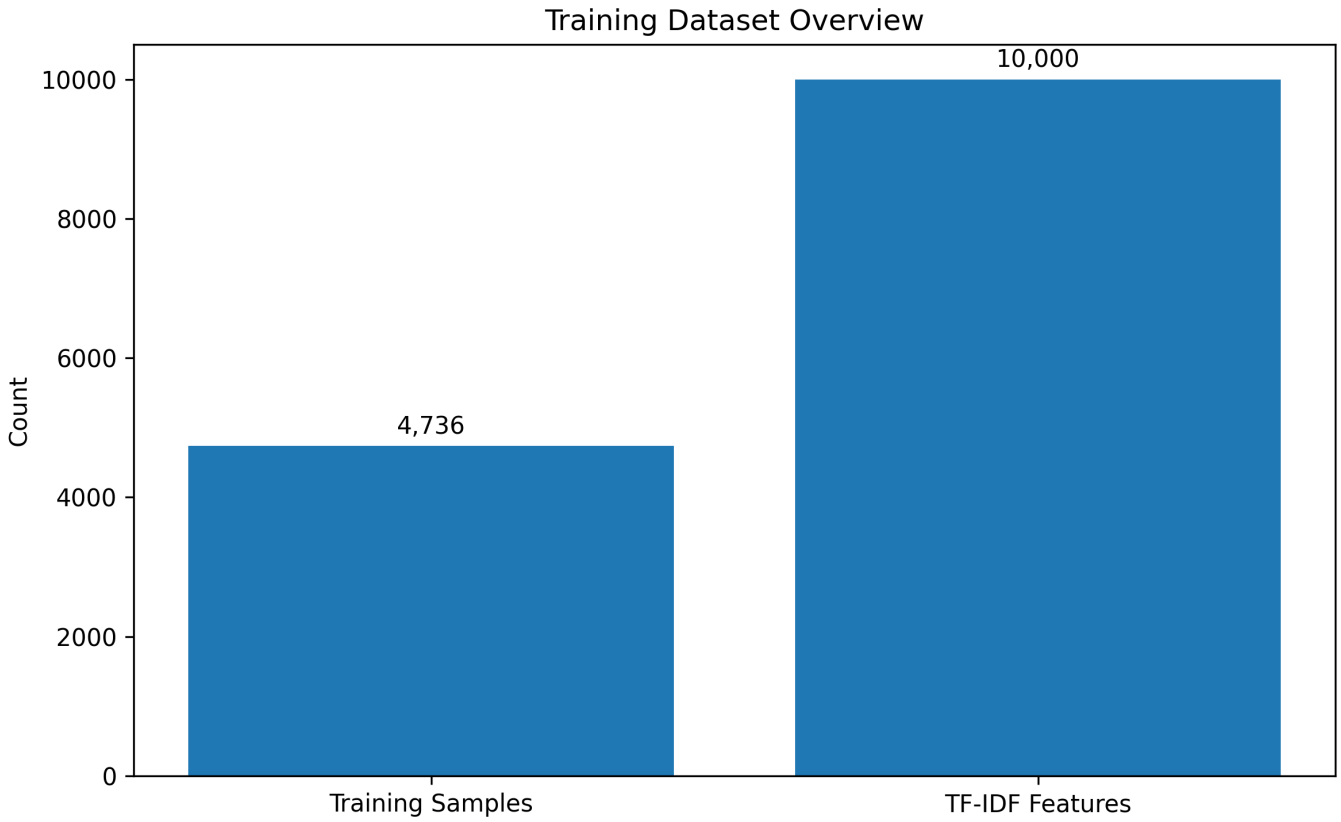
TRAINING

training_configuration

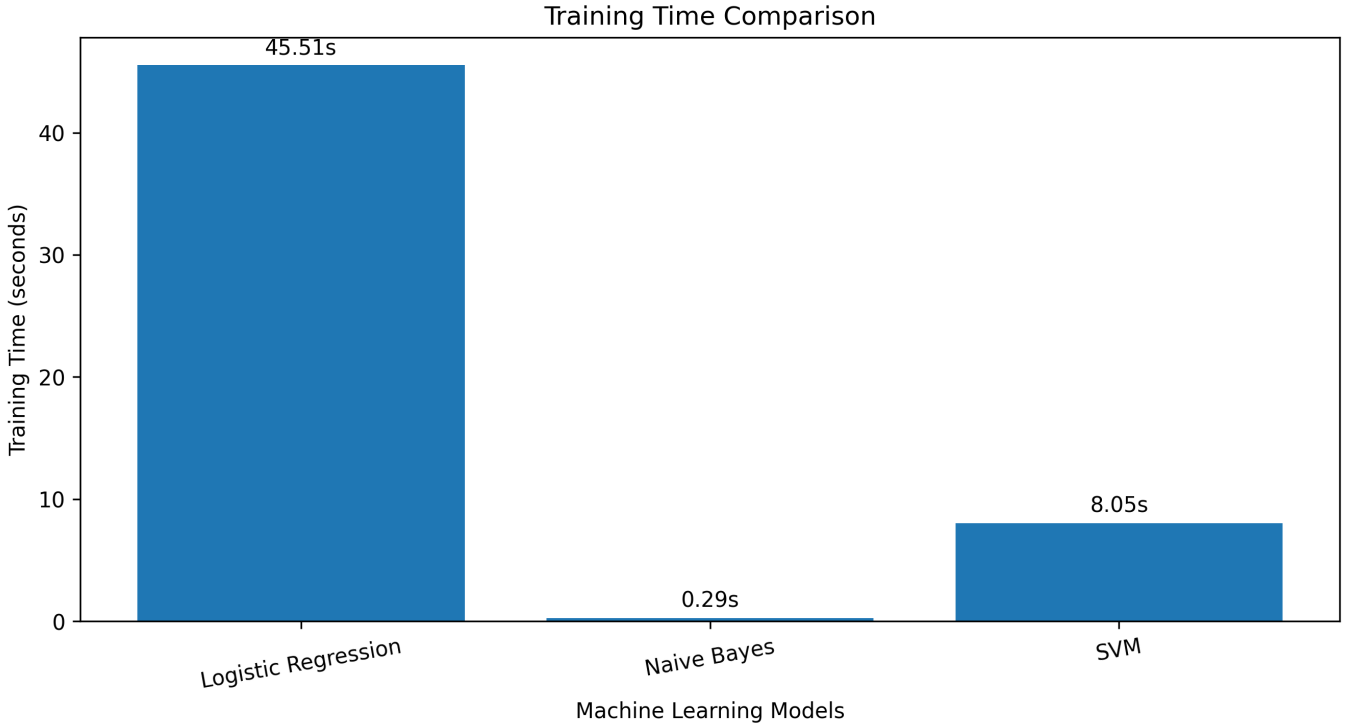
Model Training Configuration

Model	Learning Rate	Iterations	Lambda
Logistic Regression	0.1	1000	N/A
Naive Bayes	N/A	N/A	N/A
SVM	0.01	100	0.01

training_dataset_overview

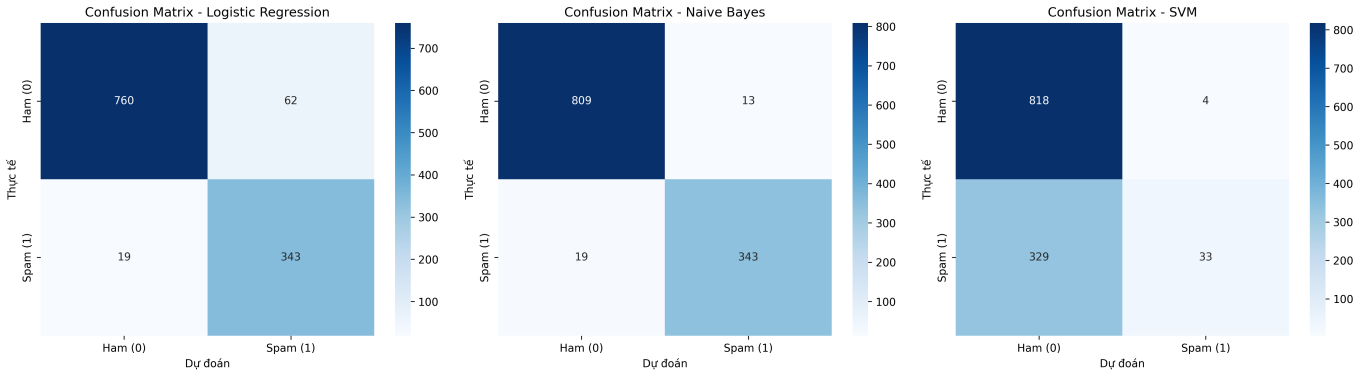


training_time_comparison

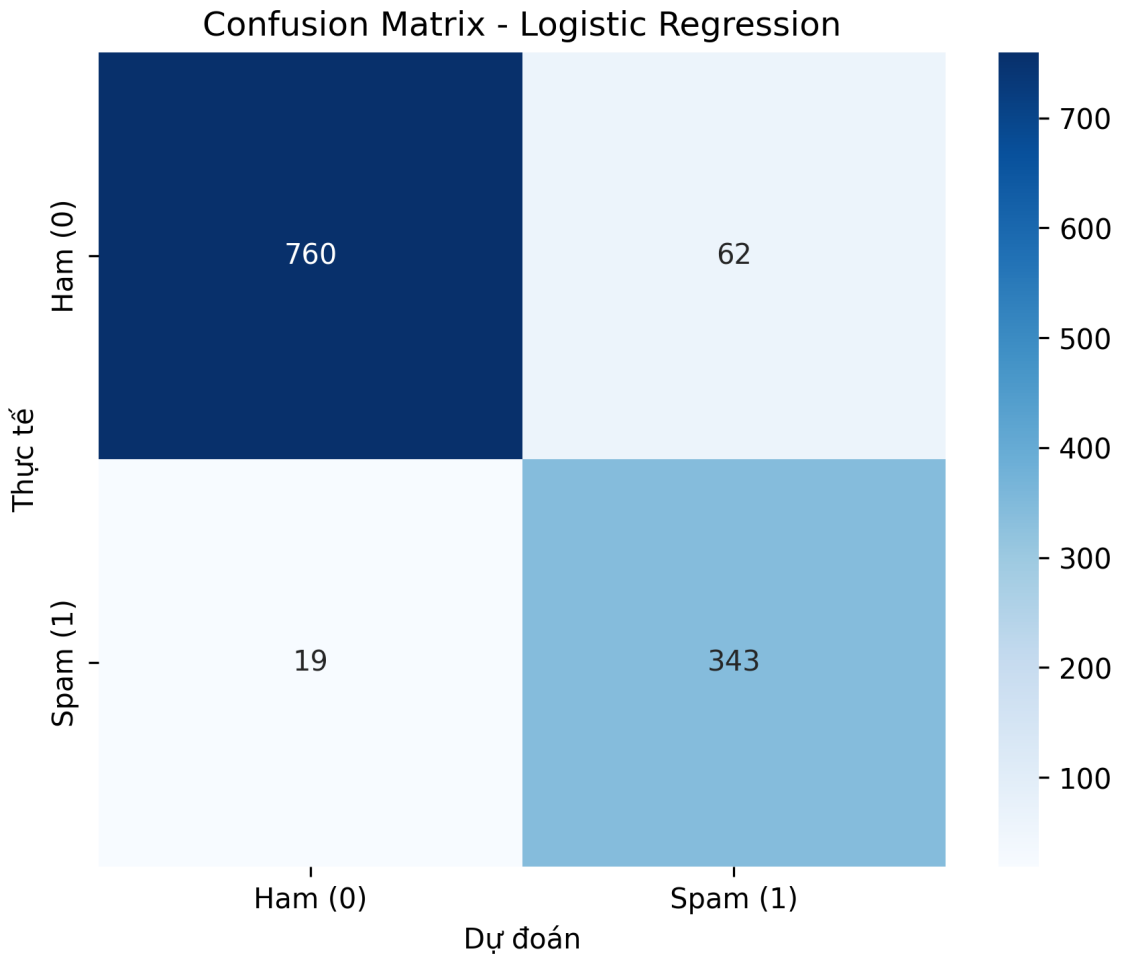


EVALUATION

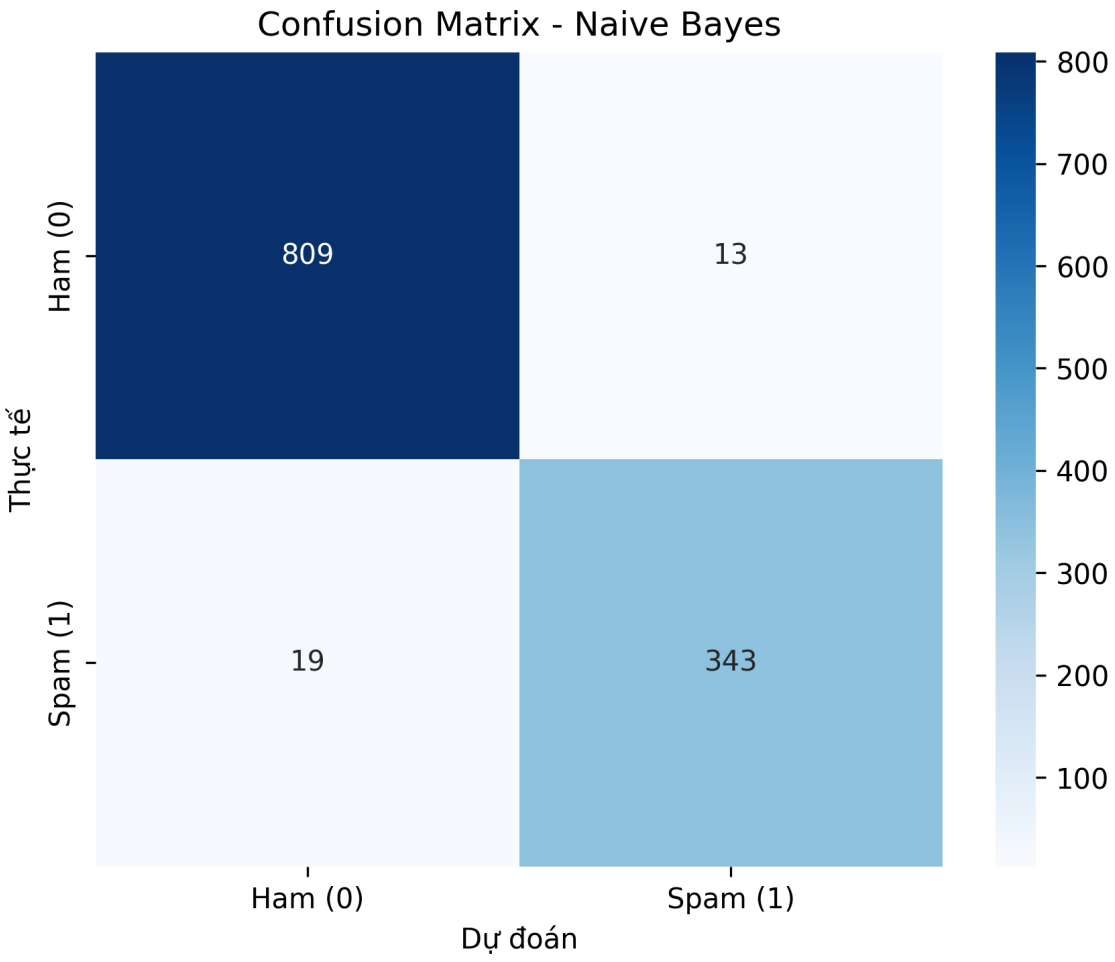
confusion_matrix_all



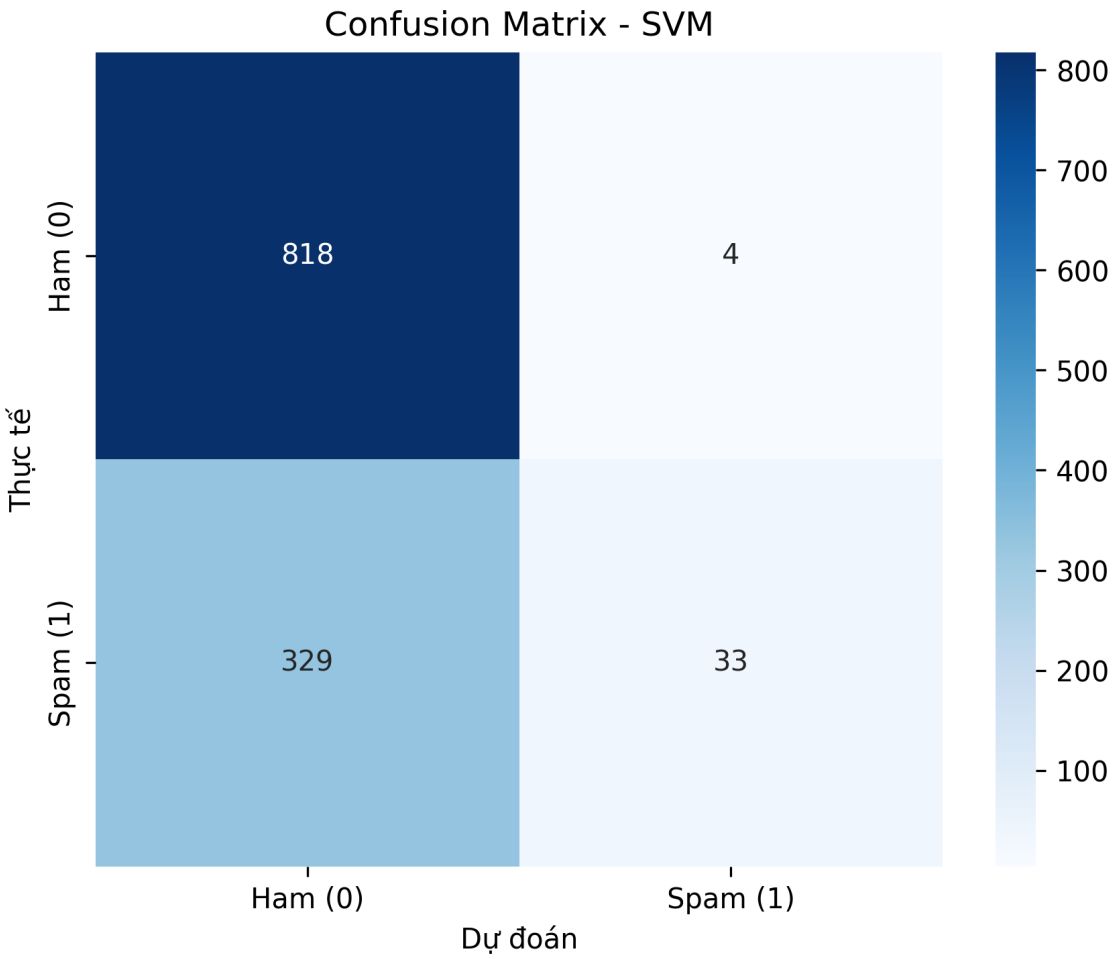
confusion_matrix_lr



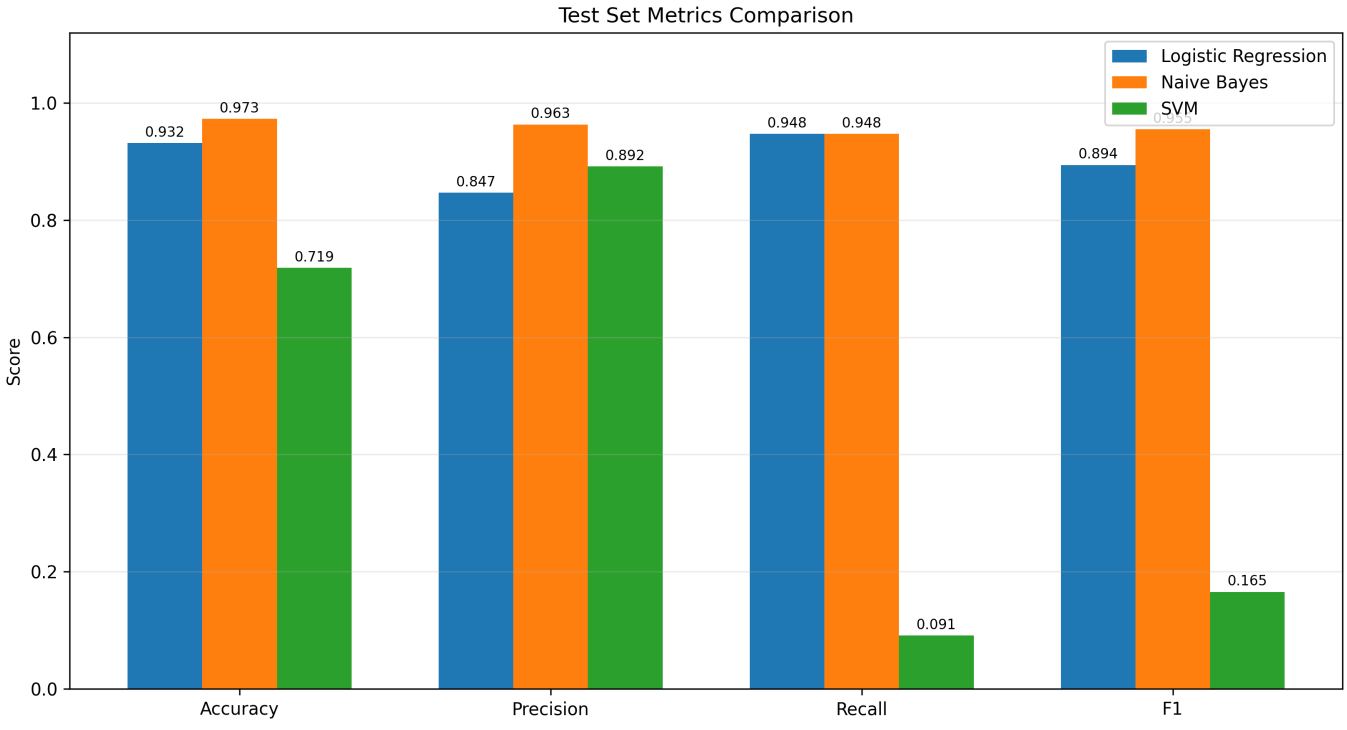
confusion_matrix_nb



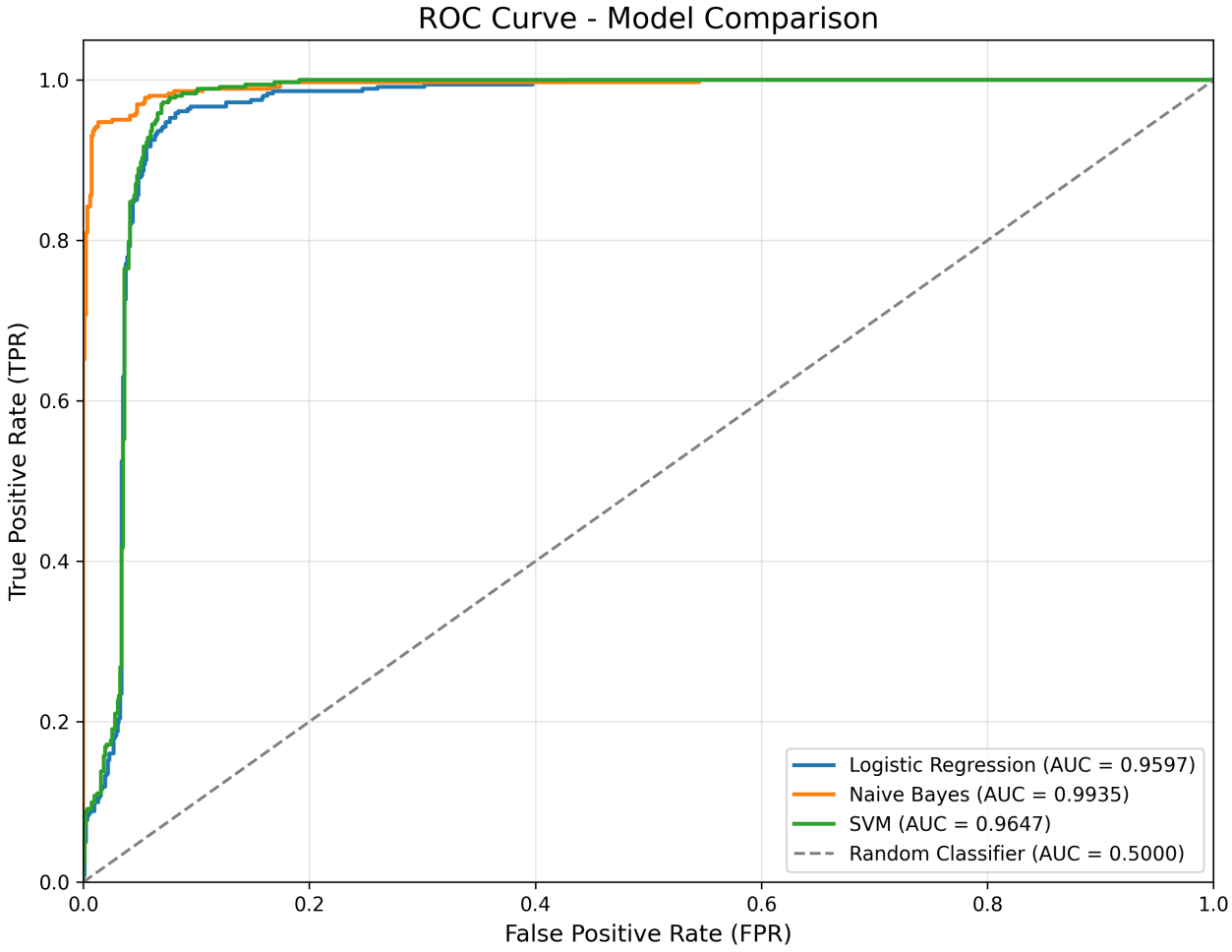
confusion_matrix_svm



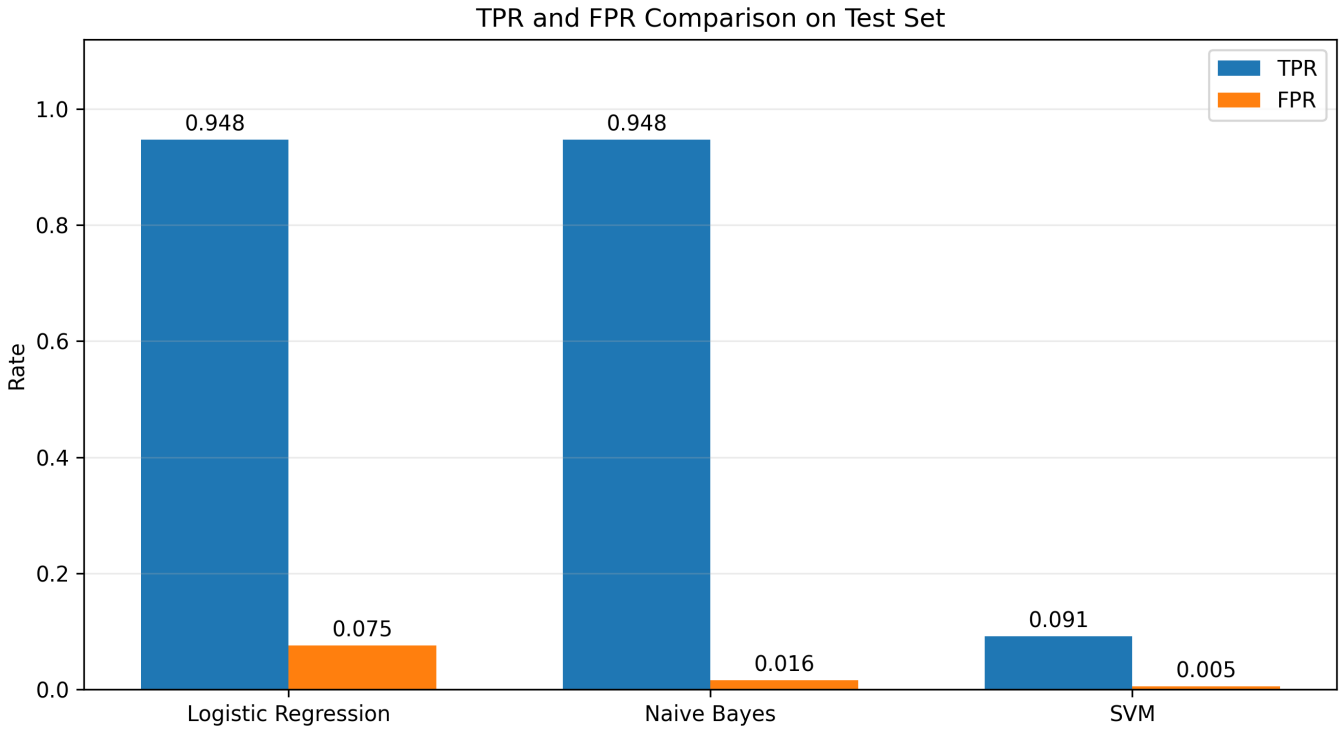
metrics_comparison



roc_curve_all



tpr_fpr_comparison



5. NHAT KY THUC THI

[2026-11-05T01:37:38+07:00] [START] [PIPELINE] Bat dau quy trinh may hoc.

[2026-11-05T01:37:38+07:00] [START] [EDA] Bat dau buoc EDA.

[2026-11-05T01:37:38+07:00] [START] [EDA] Bat dau phan tich du lieu

[2026-11-05T01:37:39+07:00] [SUCCESS] [EDA] Hoan thanh EDA trong 0.5605 giay

[2026-11-05T01:37:39+07:00] [SUCCESS] [EDA] Hoan thanh trong 0.58 giay.

[2026-11-05T01:37:39+07:00] [START] [PREPROCESSING] Bat dau buoc PREPROCESSING.

[2026-11-05T01:37:39+07:00] [START] [PREPROCESSING] Bat dau tien xu ly du lieu

[2026-11-05T01:37:43+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh tien xu ly trong 3.8277 giay

[2026-11-05T01:37:43+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh trong 3.85 giay.

[2026-11-05T01:37:43+07:00] [START] [TRAINING] Bat dau buoc TRAINING.

[2026-11-05T01:37:43+07:00] [START] [TRAINING] Bat dau huan luyen mo hinh

[2026-11-05T01:38:28+07:00] [SUCCESS] [TRAINING - Logistic Regression] Training completed in 45.51 seconds

[2026-11-05T01:38:29+07:00] [SUCCESS] [TRAINING - Naive Bayes] Training completed in 0.29 seconds

[2026-11-05T01:38:37+07:00] [SUCCESS] [TRAINING - SVM] Training completed in 8.05 seconds

[2026-11-05T01:38:37+07:00] [SUCCESS] [TRAINING] Hoan thanh training trong 54.3011 giay

[2026-11-05T01:38:37+07:00] [SUCCESS] [TRAINING] Hoan thanh trong 54.34 giay.

[2026-11-05T01:38:37+07:00] [START] [EVALUATION] Bat dau buoc EVALUATION.

[2026-11-05T01:38:37+07:00] [START] [EVALUATION] Bat dau danh gia cac mo hinh.

[2026-11-05T01:38:37+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Train] Danh gia Logistic Regression tren Train hoan tat. F1=0.9262, AUC=0.9716

[2026-11-05T01:38:37+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Test] Danh gia Logistic Regression tren Test hoan tat. F1=0.8944, AUC=0.9597

[2026-11-05T01:38:38+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Train] Danh gia Naive Bayes tren Train hoan tat. F1=0.9938, AUC=1.0000

[2026-11-05T01:38:38+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Test] Danh gia Naive Bayes tren Test hoan tat. F1=0.9554, AUC=0.9935

[2026-11-05T01:38:38+07:00] [SUCCESS] [EVALUATION - SVM - Train] Danh gia SVM tren Train hoan tat. F1=0.1699, AUC=0.9748

[2026-11-05T01:38:39+07:00] [SUCCESS] [EVALUATION - SVM - Test] Danh gia SVM tren Test hoan tat. F1=0.1654, AUC=0.9647

[2026-11-05T01:38:40+07:00] [SUCCESS] [EVALUATION] Danh gia hoan tat trong 3.24 giay.

[2026-11-05T01:38:40+07:00] [SUCCESS] [EVALUATION] Hoan thanh trong 3.26 giay.

[2026-11-05T01:38:40+07:00] [SUCCESS] [PIPELINE] Hoan thanh toan bo quy trinh trong 62.03 giay.